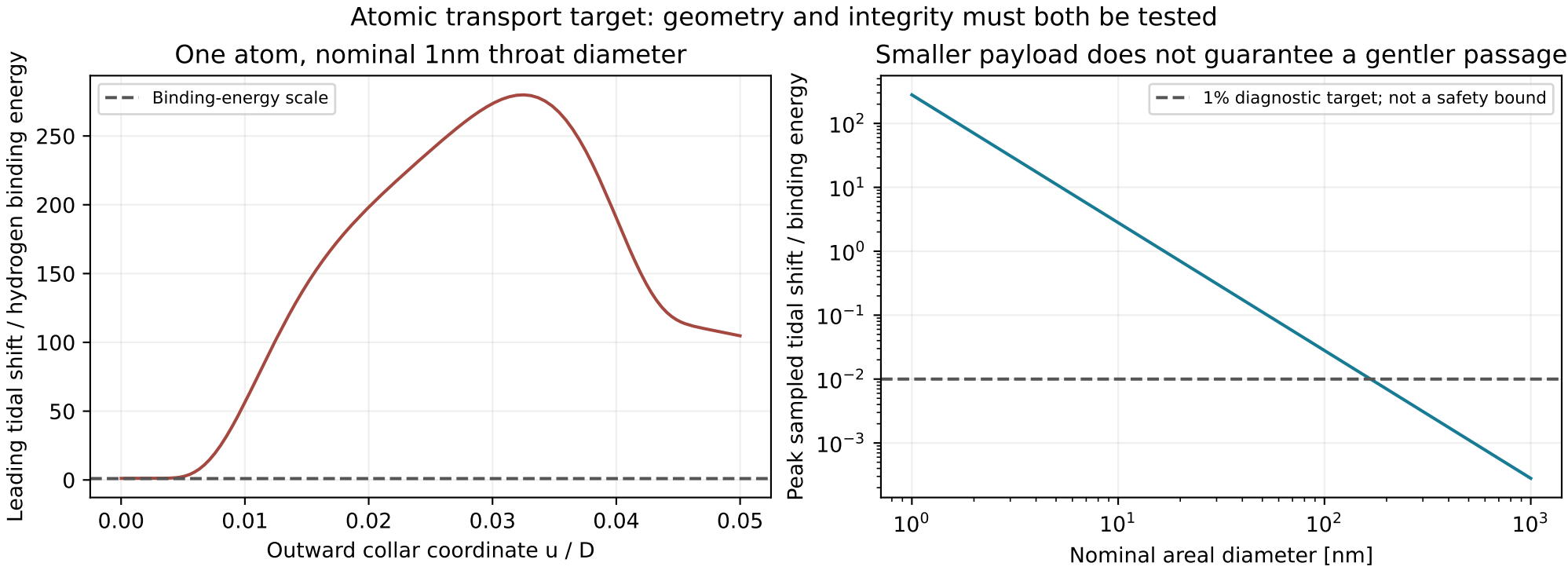
Atomic transport target: geometry and integrity must both be tested
One atom, nominal 1nm throat diameter
Binding-energy scale
Leading tidal shift / hydrogen binding energy
250
200
150
100
50
0
0.00
0.01
0.02
0.03
0.04
0.05
Outward collar coordinate u / D
Smaller payload does not guarantee a gentler passage
1% diagnostic target; not a safety bound
Peak sampled tidal shift / binding energy
$10^{2}$
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{0}$
$10^{1}$
$10^{2}$
$10^{3}$
Nominal areal diameter [nm]
No atom transit is simulated. Curves use a local instantaneous tidal diagnostic; the supporting field remains unknown.